Bangladesh University of Engineering and Technology
DEPARTMENT OF BIOMEDICAL ENGINEERING

Course Outline

Course No: BME 203
Course Title: Human Physiology
Credit hours: 3
Contact hours: 3
Level/Term: Level 2, Term 1
Academic Term: July 2025

Course Teacher(s)

Name	Office/Room	Email
Ms. Chaity Chakraborty	ECE-1013	chaity@bme.buet.ac.bd
Ms. Tasnia Binte Mamun	ECE-1125	

Course Contents

Cell physiology; membrane transport; nerve and muscle; cardiovascular, respiratory and renal physiology.

Course Objectives

- Introduce physiology
- Relate function to disease

Course Outcomes (COs)

CO No.	CO Statement	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
1	Explain human physiological mechanisms by applying basic principles of biology and chemistry	PO1	C2	Lectures, Homework	Written exams
2	Describe the fundamental mechanisms underlying normal function of cells, tissues, organs, and organ systems in humans	PO1	C1	Lectures	Written exams
4	Apply knowledge of functional mechanisms and their regulation to explain the pathophysiology of common diseases	PO1	C3	Lectures	Written exams

CO-PO Mapping

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓											
CO2	✓											
CO4	✓											

Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

COs	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
CO-1	✓	✓																		
CO-2	✓	✓																		
CO-4	✓	✓	✓	✓					✓											

* PO1: Engineering knowledge; PO2: Problem analysis; PO3: Design/development of solutions; PO4: Investigation; PO5: Modern tool usage; PO6: Engineer and society; PO7: Environment and sustainability; PO8: Ethics; PO9: Individual and teamwork; PO10: Communication; PO11: Project

management and finance; PO12: Lifelong learning.

** Domains: C = Cognitive (C1 Remember … C6 Create), PS = Psychomotor (PS1–PS7), AF = Affective (AF1–AF5). K1–K8: knowledge profile; P1–P7: range of complex engineering problem solving; A1–A5: range of complex engineering activities (BAETE).

Assessment Strategy

Class participation and attendance are recorded in every class. Continuous assessment through class tests. A comprehensive term final at the end of the term.

Distribution of Marks

Component	Share
Class participation	10%
Continuous assessment	20%
Final examination	70%
Total	**100%**

Textbooks and References

- Silverthorn, Human Physiology: An Integrated Approach, 7th ed.
- Guyton and Hall, Textbook of Medical Physiology, 13th ed.

Lecture Plan

Lecture(s)	Topics	CO(s)
1-3	Cell physiology	CO1, CO2
4-6	Membrane transport	CO1

Prepared by

Ms. Chaity Chakraborty
Name, signature and date

Ms. Tasnia Binte Mamun
Name, signature and date